

SNx4AHCT86 Quadruple 2-Input Exclusive-OR Gates

1 Features

- Inputs are TTL-Voltage Compatible
- Latch-Up Performance Exceeds 250 mA Per JESD 17
- On Products Compliant to MIL-PRF-38535, All Parameters are Tested Unless Otherwise Noted. On All Other Products, Production Processing Does Not Necessarily Include Testing of All Parameters.
- ESD Protection Exceeds JESD 22
  - 2000-V Human-Body Model (A114-A)
  - 200-V Machine Model (A115-A)

2 Applications

- Server
- PCs and Notebooks
- Network Switches
- Wearable Health and Fitness Devices
- Telecom Infrastructures
- Electronic Points of Sale

4 Simplified Schematic

An exclusive-OR gate has many applications, some of which can be represented better by alternative logic symbols.



These are five equivalent exclusive-OR symbols valid for an SN74AHCT86 gate in positive logic; negation may be shown at any two ports.

LOGIC-IDENTITY ELEMENT



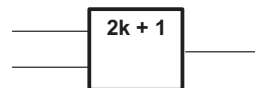
The output is active (low) if all inputs stand at the same logic level (that is,  $A = B$ ).

EVEN-PARITY ELEMENT



The output is active (low) if an even number of inputs (that is, 0 or 2) are active.

ODD-PARITY ELEMENT



The output is active (high) if an odd number of inputs (that is, only 1 of the 2) are active.

3 Description

The SNx4AHCT86 devices are quadruple 2-input exclusive-OR gates. These devices perform the Boolean function  $Y = A \times B$  or  $Y = \overline{A}B + A\overline{B}$  in positive logic.

Device Information<sup>(1)</sup>

| PART NUMBER | PACKAGE    | BODY SIZE (NOM)    |
|-------------|------------|--------------------|
| SNx4AHCT574 | TVSOP (14) | 3.60 mm × 4.40 mm  |
|             | SOIC (14)  | 8.65 mm × 3.91 mm  |
|             | SOP (14)   | 10.30 mm × 5.30 mm |
|             | SSOP (14)  | 6.20 mm × 5.30 mm  |
|             | TSSOP (14) | 5.00 mm × 4.40 mm  |

(1) For all available packages, see the orderable addendum at the end of the data sheet.

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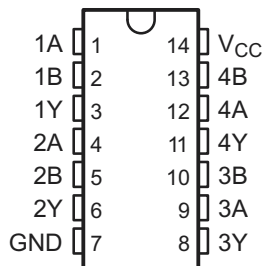
## 5 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

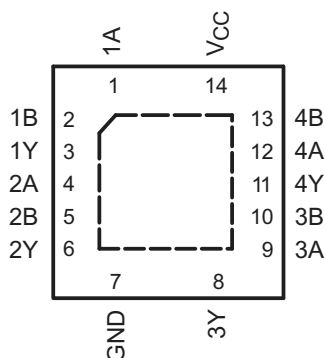
| Changes from Revision M (July 2003) to Revision N                                            | Page |
|----------------------------------------------------------------------------------------------|------|
| • Updated document to new TI data sheet format .....                                         | 1    |
| • Deleted Ordering Information table. ....                                                   | 1    |
| • Added Military Disclaimer to Features list .....                                           | 1    |
| • Added Applications. ....                                                                   | 1    |
| • Added Pin Functions table.....                                                             | 3    |
| • Added Handling Ratings table. ....                                                         | 4    |
| • Changed MAX operating temperature to 125°C in Recommended Operating Conditions table. .... | 4    |
| • Added Thermal Information table. ....                                                      | 5    |
| • Added –40°C to 125°C for SN74AHCT86 in the Electrical Characteristics table.....           | 5    |
| • Added –40°C to 125°C for SN74AHCT86 in the Switching Characteristics table.....            | 5    |
| • Added Typical Characteristics. ....                                                        | 6    |
| • Added Detailed Description section.....                                                    | 8    |
| • Added Application and Implementation section.....                                          | 9    |
| • Added Power Supply Recommendations and Layout sections.....                                | 10   |

## 6 Pin Configuration and Functions

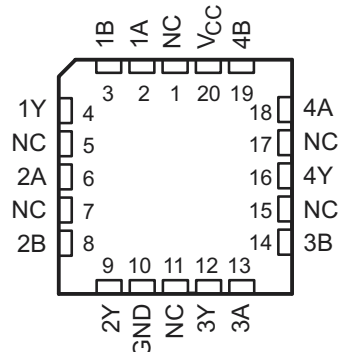
**SN54AHCT86 . . . J OR W PACKAGE  
SN74AHCT86 . . . D, DB, DGV, N, NS,  
OR PW PACKAGE  
(TOP VIEW)**



**SN74AHCT86 . . . RGY PACKAGE  
(TOP VIEW)**



**SN54AHCT86 . . . FK PACKAGE  
(TOP VIEW)**



NC – No internal connection

### Pin Functions

| PIN             |                       |     |            |    | I/O | DESCRIPTION   |
|-----------------|-----------------------|-----|------------|----|-----|---------------|
| NAME            | SN74AHCT86            |     | SN54AHCT86 |    |     |               |
|                 | D, DB, DGV, N, NS, PW | RGY | J, W       | FK |     |               |
| 1A              | 1                     | 1   | 1          | 2  | I   | 1A Input      |
| 1B              | 2                     | 2   | 2          | 3  | I   | 1B Input      |
| 1Y              | 3                     | 3   | 3          | 4  | O   | 1Y Output     |
| 2A              | 4                     | 4   | 4          | 6  | I   | 2A Input      |
| 2B              | 5                     | 5   | 5          | 8  | I   | 2B Input      |
| 2Y              | 6                     | 6   | 6          | 9  | O   | 2Y Output     |
| 3Y              | 8                     | 8   | 8          | 12 | O   | 3Y Output     |
| 3A              | 9                     | 9   | 9          | 13 | I   | 3A Input      |
| 3B              | 10                    | 10  | 10         | 14 | I   | 3B Input      |
| 4Y              | 11                    | 11  | 11         | 16 | O   | 4Y Output     |
| 4A              | 12                    | 12  | 12         | 18 | I   | 4A Input      |
| 4B              | 13                    | 13  | 13         | 19 | I   | 4B Input      |
| GND             | 7                     | 7   | 7          | 10 | —   | Ground Pin    |
| NC              | —                     | —   | —          | 1  | —   | No Connection |
|                 |                       |     |            | 5  |     |               |
|                 |                       |     |            | 7  |     |               |
|                 |                       |     |            | 11 |     |               |
|                 |                       |     |            | 15 |     |               |
|                 |                       |     |            | 17 |     |               |
| V <sub>CC</sub> | 14                    | 14  | 14         | 20 | —   | Power Pin     |

## 7 Specifications

### 7.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)<sup>(1)</sup>

|          |                                            | MIN                         | MAX            | UNIT |
|----------|--------------------------------------------|-----------------------------|----------------|------|
| $V_{CC}$ | Supply voltage range                       | −0.5                        | 7              | V    |
| $V_I$    | Input voltage range <sup>(2)</sup>         | −0.5                        | 7              | V    |
| $V_O$    | Output voltage range <sup>(2)</sup>        | −0.5                        | $V_{CC} + 0.5$ | V    |
| $I_{IK}$ | Input clamp current                        | $V_I < 0$                   | −20            | mA   |
| $I_{OK}$ | Output clamp current                       | $V_O < 0$ or $V_O > V_{CC}$ | ±20            | mA   |
| $I_O$    | Continuous output current                  | $V_O = 0$ to $V_{CC}$       | ±25            | mA   |
|          | Continuous current through $V_{CC}$ or GND |                             | ±50            | mA   |

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions* is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
- (2) The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

### 7.2 Handling Ratings

|             |                                                                                          | MIN | MAX  | UNIT |
|-------------|------------------------------------------------------------------------------------------|-----|------|------|
| $T_{stg}$   | Storage temperature range                                                                | −65 | 150  | °C   |
| $V_{(ESD)}$ | Electrostatic discharge                                                                  |     |      |      |
|             | Human body model (HBM), per ANSI/ESDA/JEDEC JS-001, all pins <sup>(1)</sup>              | 0   | 2000 | V    |
|             | Charged device model (CDM), per JEDEC specification JESD22-C101, all pins <sup>(2)</sup> | 0   | 1000 |      |

- (1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
- (2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

### 7.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)<sup>(1)</sup>

|                     |                                    | SN54AHCT86 |          | SN74AHCT86 |          | UNIT |
|---------------------|------------------------------------|------------|----------|------------|----------|------|
|                     |                                    | MIN        | MAX      | MIN        | MAX      |      |
| $V_{CC}$            | Supply voltage                     | 4.5        | 5.5      | 4.5        | 5.5      | V    |
| $V_{IH}$            | High-level input voltage           | 2          |          | 2          |          | V    |
| $V_{IL}$            | Low-level input voltage            |            | 0.8      |            | 0.8      | V    |
| $V_I$               | Input voltage                      | 0          | 5.5      | 0          | 5.5      | V    |
| $V_O$               | Output voltage                     | 0          | $V_{CC}$ | 0          | $V_{CC}$ | V    |
| $I_{OH}$            | High-level output current          |            | −8       |            | −8       | mA   |
| $I_{OL}$            | Low-level output current           |            | 8        |            | 8        | mA   |
| $\Delta t/\Delta v$ | Input transition rise or fall rate |            | 20       |            | 20       | ns/V |
| $T_A$               | Operating free-air temperature     | −55        | 125      | −40        | 125      | °C   |

- (1) All unused inputs of the device must be held at  $V_{CC}$  or GND to ensure proper device operation. Refer to the TI application report, *Implications of Slow or Floating CMOS Inputs* (SCBA004).

## 7.4 Thermal Information

| THERMAL METRIC <sup>(1)</sup> |                                              | SN74AHCT86 |       |       |      |      |       |      | UNIT |
|-------------------------------|----------------------------------------------|------------|-------|-------|------|------|-------|------|------|
|                               |                                              | D          | DB    | DGV   | N    | NS   | PW    | RGY  |      |
|                               |                                              | 14 PINS    |       |       |      |      |       |      |      |
| R <sub>θJA</sub>              | Junction-to-ambient thermal resistance       | 97.5       | 109.5 | 133.3 | 59.7 | 92.2 | 125.1 | 59.0 | °C/W |
| R <sub>θJC(top)</sub>         | Junction-to-case (top) thermal resistance    | 58.7       | 62.1  | 55.6  | 47.3 | 49.8 | 53.7  | 72.5 |      |
| R <sub>θJB</sub>              | Junction-to-board thermal resistance         | 51.8       | 56.9  | 66.3  | 39.5 | 51.0 | 66.9  | 35.0 |      |
| ψ <sub>JT</sub>               | Junction-to-top characterization parameter   | 22.6       | 22.6  | 7.8   | 32.4 | 15.7 | 7.6   | 3.9  |      |
| ψ <sub>JB</sub>               | Junction-to-board characterization parameter | 51.6       | 56.3  | 56.6  | 39.4 | 50.6 | 66.3  | 35.1 |      |
| R <sub>θJC(bot)</sub>         | Junction-to-case (bottom) thermal resistance | n/a        | n/a   | n/a   | n/a  | n/a  | n/a   | 15.4 |      |

(1) For more information about traditional and new thermal metrics, see the *IC Package Thermal Metrics* application report ([SPRA953](#)).

## 7.5 Electrical Characteristics

over recommended operating free-air temperature range (unless otherwise noted)

| PARAMETER                       | TEST CONDITIONS                                               | V <sub>CC</sub> | T <sub>A</sub> = 25°C |     |      | SN54AHCT86 |                   | –40°C to 85°C<br>SN74AHCT86 |      | –40°C to 125°C<br>SN74AHCT86 |      | UNIT |
|---------------------------------|---------------------------------------------------------------|-----------------|-----------------------|-----|------|------------|-------------------|-----------------------------|------|------------------------------|------|------|
|                                 |                                                               |                 | MIN                   | TYP | MAX  | MIN        | MAX               | MIN                         | MAX  | MIN                          | MAX  |      |
| V <sub>OH</sub>                 | I <sub>OH</sub> = –50 μA                                      | 4.5 V           | 4.4                   | 4.5 |      | 4.4        |                   | 4.4                         |      | 4.4                          |      | V    |
|                                 | I <sub>OH</sub> = –8 mA                                       |                 | 3.94                  |     |      | 3.8        |                   | 3.8                         |      | 3.8                          |      |      |
| V <sub>OL</sub>                 | I <sub>OL</sub> = 50 μA                                       | 4.5 V           |                       |     | 0.1  |            | 0.1               |                             | 0.1  |                              | 0.1  | V    |
|                                 | I <sub>OL</sub> = 8 mA                                        |                 |                       |     | 0.36 |            | 0.44              |                             | 0.44 |                              | 0.44 |      |
| I <sub>I</sub>                  | V <sub>I</sub> = 5.5 V or GND                                 | 0 V to 5.5 V    |                       |     | ±0.1 |            | ±1 <sup>(1)</sup> |                             | ±1   |                              | ±1   | μA   |
| I <sub>CC</sub>                 | V <sub>I</sub> = V <sub>CC</sub> or GND, I <sub>O</sub> = 0   | 5.5 V           |                       |     | 2    |            | 20                |                             | 20   |                              | 20   | μA   |
| ΔI <sub>CC</sub> <sup>(2)</sup> | One input at 3.4 V,<br>Other inputs at V <sub>CC</sub> or GND | 5.5 V           |                       |     | 1.35 |            | 1.5               |                             | 1.5  |                              | 1.5  | mA   |
| C <sub>i</sub>                  | V <sub>I</sub> = V <sub>CC</sub> or GND                       | 5 V             |                       |     | 4 10 |            |                   |                             | 10   |                              |      | pF   |

(1) On products compliant to MIL-PRF-38535, this parameter is not production tested at V<sub>CC</sub> = 0 V.

(2) This is the increase in supply current for each input at one of the specified TTL voltage levels, rather than 0 V or V<sub>CC</sub>.

## 7.6 Switching Characteristics

over recommended operating free-air temperature range, V<sub>CC</sub> = 5 V ± 0.5 V (unless otherwise noted) (see [Figure 2](#))

| PARAMETER        | FROM<br>(INPUT) | TO<br>(OUTPUT) | LOAD<br>CAPACITANCE    | T <sub>A</sub> = 25°C |                  |                    | –55°C to 125°C<br>SN54AHCT86 |                  | –40°C to 85°C<br>SN74AHCT86 |     | –40°C to 125°C<br>SN74AHCT86 |     | UNIT |
|------------------|-----------------|----------------|------------------------|-----------------------|------------------|--------------------|------------------------------|------------------|-----------------------------|-----|------------------------------|-----|------|
|                  |                 |                |                        | MIN                   | TYP              | MAX                | MIN                          | MAX              | MIN                         | MAX | MIN                          | MAX |      |
| t <sub>PLH</sub> | A or B          | Y              | C <sub>L</sub> = 15 pF |                       | 5 <sup>(1)</sup> | 6.9 <sup>(1)</sup> | 1 <sup>(1)</sup>             | 8 <sup>(1)</sup> | 1                           | 8   | 1                            | 9   | ns   |
| t <sub>PHL</sub> |                 |                |                        |                       | 5 <sup>(1)</sup> | 6.9 <sup>(1)</sup> | 1 <sup>(1)</sup>             | 8 <sup>(1)</sup> | 1                           | 8   | 1                            | 9   |      |
| t <sub>PLH</sub> | A or B          | Y              | C <sub>L</sub> = 50 pF |                       | 5.5              | 8.8                | 1                            | 10               | 1                           | 9   | 1                            | 11  | ns   |
| t <sub>PHL</sub> |                 |                |                        |                       | 5.5              | 8.8                | 1                            | 10               | 1                           | 9   | 1                            | 11  |      |

(1) On products compliant to MIL-PRF-38535, this parameter is not production tested.

## 7.7 Noise Characteristics

 $V_{CC} = 5\text{ V}$ ,  $C_L = 50\text{ pF}$ ,  $T_A = 25^\circ\text{C}^{(1)}$ 

| PARAMETER   |                                        | SN74AHCT86 |      |      | UNIT |
|-------------|----------------------------------------|------------|------|------|------|
|             |                                        | MIN        | TYP  | MAX  |      |
| $V_{OL(P)}$ | Quiet output, maximum dynamic $V_{OL}$ |            | 0.4  | 0.8  | V    |
| $V_{OL(V)}$ | Quiet output, minimum dynamic $V_{OL}$ |            | −0.4 | −0.8 | V    |
| $V_{OH(V)}$ | Quiet output, minimum dynamic $V_{OH}$ | 4.4        |      |      | V    |
| $V_{IH(D)}$ | High-level dynamic input voltage       | 2          |      |      | V    |
| $V_{IL(D)}$ | Low-level dynamic input voltage        |            |      | 0.8  | V    |

(1) Characteristics are for surface-mount packages only.

## 7.8 Operating Characteristics

 $V_{CC} = 5\text{ V}$ ,  $T_A = 25^\circ\text{C}$ 

| PARAMETER |                               | TEST CONDITIONS |                    | TYP | UNIT |
|-----------|-------------------------------|-----------------|--------------------|-----|------|
| $C_{pd}$  | Power dissipation capacitance | No load,        | $f = 1\text{ MHz}$ | 18  | pF   |

## 7.9 Typical Characteristics

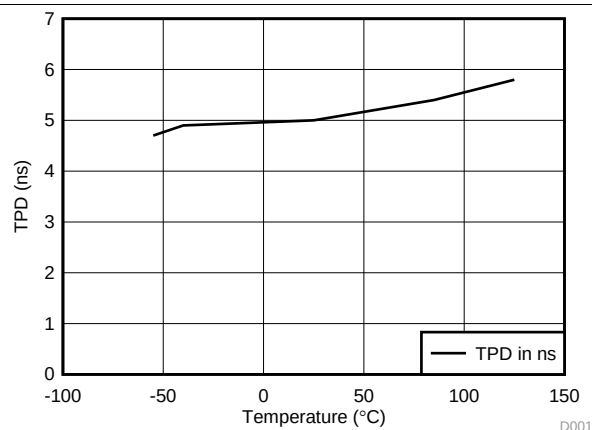
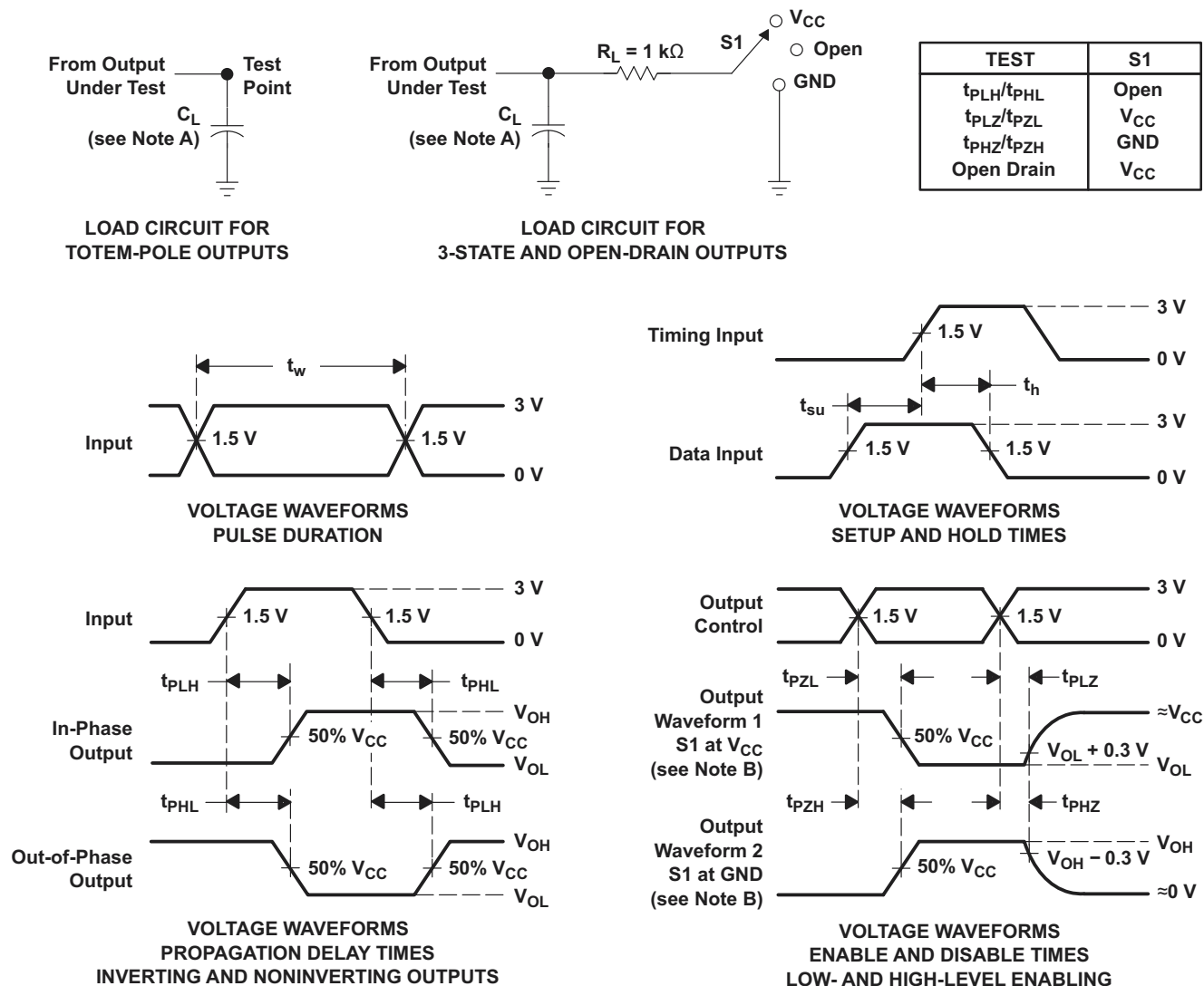


Figure 1. TPD vs Temperature

## 8 Parameter Measurement Information



- NOTES: A.  $C_L$  includes probe and jig capacitance.
- B. Waveform 1 is for an output with internal conditions such that the output is low except when disabled by the output control. Waveform 2 is for an output with internal conditions such that the output is high except when disabled by the output control.
- C. All input pulses are supplied by generators having the following characteristics:  $PRR \leq 1\text{ MHz}$ ,  $Z_O = 50\ \Omega$ ,  $t_r \leq 3\text{ ns}$ ,  $t_f \leq 3\text{ ns}$ .
- D. The outputs are measured one at a time with one input transition per measurement.
- E. All parameters and waveforms are not applicable to all devices.

**Figure 2. Load Circuit and Voltage Waveforms**

## 9 Detailed Description

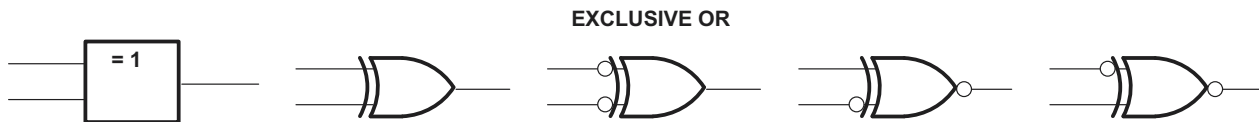
### 9.1 Overview

The SNx4AHCT86 devices are quadruple 2-input exclusive-OR gates. These devices perform the Boolean function  $Y = A \times B$  or  $Y = \overline{A}B + A\overline{B}$  in positive logic.

The inputs are TTL compatible allowing 3.3 V to 5 V translation.

### 9.2 Functional Block Diagram

An exclusive-OR gate has many applications, some of which can be represented better by alternative logic symbols.



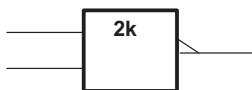
These are five equivalent exclusive-OR symbols valid for an SN74AHCT86 gate in positive logic; negation may be shown at any two ports.

#### LOGIC-IDENTITY ELEMENT



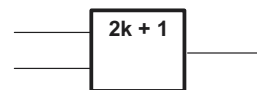
The output is active (low) if all inputs stand at the same logic level (that is,  $A = B$ ).

#### EVEN-PARITY ELEMENT



The output is active (low) if an even number of inputs (that is, 0 or 2) are active.

#### ODD-PARITY ELEMENT



The output is active (high) if an odd number of inputs (that is, only 1 of the 2) are active.

**Figure 3. Exclusive-OR Logic**

### 9.3 Feature Description

- TTL inputs
  - Lowered switching threshold allows up translation 3.3 V to 5 V
- Slow edges reduce output ringing

### 9.4 Device Functional Modes

**Table 1. Function Table  
(Each Gate)**

| INPUTS |   | OUTPUT<br>Y |
|--------|---|-------------|
| A      | B |             |
| L      | L | L           |
| L      | H | H           |
| H      | L | H           |
| H      | H | L           |



## 10 Application and Implementation

### 10.1 Application Information

The SNx4AHCT86 is a low-drive CMOS device that can be used for a multitude of bus interface type applications where output ringing is a concern. The low drive and slow edge rates will minimize overshoot and undershoot on the outputs. The input switching levels have been lowered to accommodate TTL inputs of 0.8-V  $V_{IL}$  and 2-V  $V_{IH}$ . This feature makes the device ideal for translating up from 3.3 V to 5 V. Figure 5 shows this type of translation.

### 10.2 Typical Application

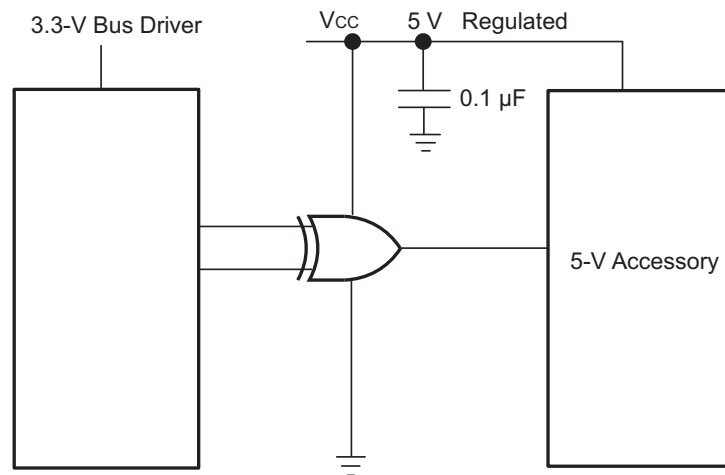


Figure 4. Typical Application Schematic

#### 10.2.1 Design Requirements

This device uses CMOS technology and has balanced output drive. Care should be taken to avoid bus contention because it can drive currents that would exceed maximum limits. The high drive will also create fast edges into light loads; therefore, routing and load conditions should be considered to prevent ringing.

#### 10.2.2 Detailed Design Procedure

1. Recommended input conditions
  - Rise time and fall time specs: See  $(\Delta t/\Delta V)$  in the [Recommended Operating Conditions](#) table.
  - Specified High and low levels: See ( $V_{IH}$  and  $V_{IL}$ ) in the [Recommended Operating Conditions](#) table.
  - Inputs are overvoltage tolerant allowing them to go as high as 5.5 V at any valid  $V_{CC}$
2. Recommend output conditions
  - Load currents should not exceed 25 mA per output and 75 mA total for the part
  - Outputs should not be pulled above  $V_{CC}$

## Typical Application (continued)

### 10.2.3 Application Curves

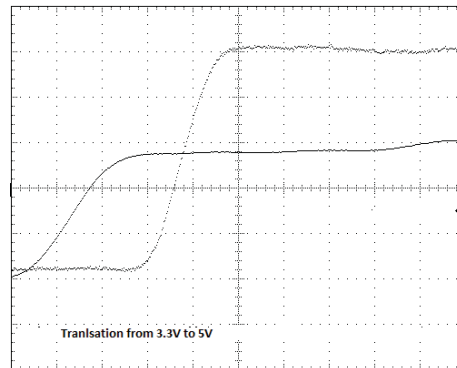


Figure 5. Up Translation

## 11 Power Supply Recommendations

The power supply can be any voltage between the MIN and MAX supply voltage rating located in the [Recommended Operating Conditions](#) table.

Each  $V_{CC}$  pin should have a good bypass capacitor to prevent power disturbance. For devices with a single supply, 0.1  $\mu\text{F}$  is recommended. If there are multiple  $V_{CC}$  pins, 0.01  $\mu\text{F}$  or 0.022  $\mu\text{F}$  is recommended for each power pin. It is acceptable to parallel multiple bypass caps to reject different frequencies of noise. A 0.1  $\mu\text{F}$  and 1  $\mu\text{F}$  are commonly used in parallel. The bypass capacitor should be installed as close to the power pin as possible for best results.

## 12 Layout

### 12.1 Layout Guidelines

When using multiple bit logic devices inputs should not ever float.

In many cases, functions or parts of functions of digital logic devices are unused, for example, when only two inputs of a triple-input AND gate are used or only 3 of the 4 buffer gates are used. Such input pins should not be left unconnected because the undefined voltages at the outside connections result in undefined operational states. Specified in [Figure 6](#) are the rules that must be observed under all circumstances. All unused inputs of digital logic devices must be connected to a high or low bias to prevent them from floating. The logic level that should be applied to any particular unused input depends on the function of the device. Generally they will be tied to GND or  $V_{CC}$ ; whichever makes more sense or is more convenient. It is generally acceptable to float outputs unless the part is a transceiver. If the transceiver has an output enable pin, it will disable the outputs section of the part when asserted. This will not disable the input section of the IOs so they cannot float when disabled.

### 12.2 Layout Example

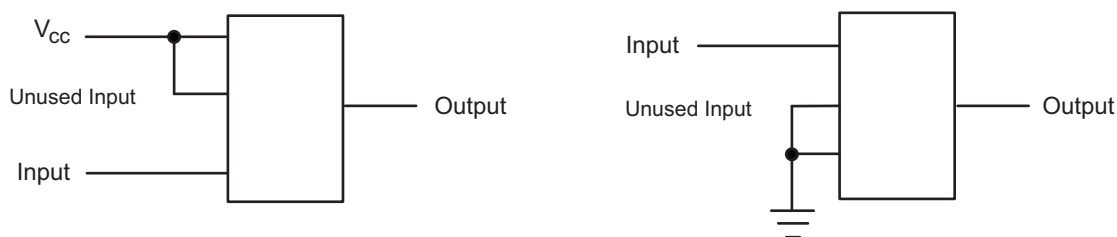


Figure 6. Layout Diagram

## 13 Device and Documentation Support

### 13.1 Related Links

The table below lists quick access links. Categories include technical documents, support and community resources, tools and software, and quick access to sample or buy.

**Table 2. Related Links**

| PARTS      | PRODUCT FOLDER             | SAMPLE & BUY               | TECHNICAL DOCUMENTS        | TOOLS & SOFTWARE           | SUPPORT & COMMUNITY        |
|------------|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|
| SN54AHCT86 | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> |
| SN74AHCT86 | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> |

### 13.2 Trademarks

All trademarks are the property of their respective owners.

### 13.3 Electrostatic Discharge Caution



These devices have limited built-in ESD protection. The leads should be shorted together or the device placed in conductive foam during storage or handling to prevent electrostatic damage to the MOS gates.

### 13.4 Glossary

[SLYZ022](#) — *TI Glossary*.

This glossary lists and explains terms, acronyms, and definitions.

## 14 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

**PACKAGING INFORMATION**

| Orderable Device | Status<br>(1) | Package Type | Package<br>Drawing | Pins | Package<br>Qty | Eco Plan<br>(2)            | Lead/Ball Finish<br>(6) | MSL Peak Temp<br>(3) | Op Temp (°C) | Device Marking<br>(4/5)                  | Samples                 |
|------------------|---------------|--------------|--------------------|------|----------------|----------------------------|-------------------------|----------------------|--------------|------------------------------------------|-------------------------|
| 5962-9681701Q2A  | ACTIVE        | LCCC         | FK                 | 20   | 1              | TBD                        | POST-PLATE              | N / A for Pkg Type   | -55 to 125   | 5962-<br>9681701Q2A<br>SNJ54AHCT<br>86FK | <a href="#">Samples</a> |
| 5962-9681701QCA  | ACTIVE        | CDIP         | J                  | 14   | 1              | TBD                        | A42                     | N / A for Pkg Type   | -55 to 125   | 5962-9681701QC<br>A<br>SNJ54AHCT86J      | <a href="#">Samples</a> |
| SN74AHCT86D      | ACTIVE        | SOIC         | D                  | 14   | 50             | Green (RoHS<br>& no Sb/Br) | CU NIPDAU               | Level-1-260C-UNLIM   | -40 to 85    | AHCT86                                   | <a href="#">Samples</a> |
| SN74AHCT86DBR    | ACTIVE        | SSOP         | DB                 | 14   | 2000           | Green (RoHS<br>& no Sb/Br) | CU NIPDAU               | Level-1-260C-UNLIM   | -40 to 125   | HB86                                     | <a href="#">Samples</a> |
| SN74AHCT86DGVR   | ACTIVE        | TVSOP        | DGV                | 14   | 2000           | Green (RoHS<br>& no Sb/Br) | CU NIPDAU               | Level-1-260C-UNLIM   | -40 to 125   | HB86                                     | <a href="#">Samples</a> |
| SN74AHCT86DGVRE4 | ACTIVE        | TVSOP        | DGV                | 14   | 2000           | Green (RoHS<br>& no Sb/Br) | CU NIPDAU               | Level-1-260C-UNLIM   | -40 to 125   | HB86                                     | <a href="#">Samples</a> |
| SN74AHCT86DR     | ACTIVE        | SOIC         | D                  | 14   | 2500           | Green (RoHS<br>& no Sb/Br) | CU NIPDAU               | Level-1-260C-UNLIM   | -40 to 125   | AHCT86                                   | <a href="#">Samples</a> |
| SN74AHCT86N      | ACTIVE        | PDIP         | N                  | 14   | 25             | Green (RoHS<br>& no Sb/Br) | CU NIPDAU               | N / A for Pkg Type   | -40 to 125   | SN74AHCT86N                              | <a href="#">Samples</a> |
| SN74AHCT86NSR    | ACTIVE        | SO           | NS                 | 14   | 2000           | Green (RoHS<br>& no Sb/Br) | CU NIPDAU               | Level-1-260C-UNLIM   | -40 to 125   | AHCT86                                   | <a href="#">Samples</a> |
| SN74AHCT86PW     | ACTIVE        | TSSOP        | PW                 | 14   | 90             | Green (RoHS<br>& no Sb/Br) | CU NIPDAU               | Level-1-260C-UNLIM   | -40 to 125   | HB86                                     | <a href="#">Samples</a> |
| SN74AHCT86PWR    | ACTIVE        | TSSOP        | PW                 | 14   | 2000           | Green (RoHS<br>& no Sb/Br) | CU NIPDAU               | Level-1-260C-UNLIM   | -40 to 125   | HB86                                     | <a href="#">Samples</a> |
| SN74AHCT86RGYR   | ACTIVE        | VQFN         | RGY                | 14   | 3000           | Green (RoHS<br>& no Sb/Br) | CU NIPDAU               | Level-2-260C-1 YEAR  | -40 to 125   | HB86                                     | <a href="#">Samples</a> |
| SN74AHCT86RGYRG4 | ACTIVE        | VQFN         | RGY                | 14   | 3000           | Green (RoHS<br>& no Sb/Br) | CU NIPDAU               | Level-2-260C-1 YEAR  | -40 to 125   | HB86                                     | <a href="#">Samples</a> |
| SNJ54AHCT86FK    | ACTIVE        | LCCC         | FK                 | 20   | 1              | TBD                        | POST-PLATE              | N / A for Pkg Type   | -55 to 125   | 5962-<br>9681701Q2A<br>SNJ54AHCT<br>86FK | <a href="#">Samples</a> |
| SNJ54AHCT86J     | ACTIVE        | CDIP         | J                  | 14   | 1              | TBD                        | A42                     | N / A for Pkg Type   | -55 to 125   | 5962-9681701QC<br>A                      | <a href="#">Samples</a> |

| Orderable Device | Status<br>(1) | Package Type | Package<br>Drawing | Pins | Package<br>Qty | Eco Plan<br>(2) | Lead/Ball Finish<br>(6) | MSL Peak Temp<br>(3) | Op Temp (°C) | Device Marking<br>(4/5) | Samples |
|------------------|---------------|--------------|--------------------|------|----------------|-----------------|-------------------------|----------------------|--------------|-------------------------|---------|
|                  |               |              |                    |      |                |                 |                         |                      |              | SNJ54AHCT86J            |         |

(1) The marketing status values are defined as follows:

**ACTIVE:** Product device recommended for new designs.

**LIFEBUY:** TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.

**NRND:** Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.

**PREVIEW:** Device has been announced but is not in production. Samples may or may not be available.

**OBsolete:** TI has discontinued the production of the device.

(2) **RoHS:** TI defines "RoHS" to mean semiconductor products that are compliant with the current EU RoHS requirements for all 10 RoHS substances, including the requirement that RoHS substance do not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, "RoHS" products are suitable for use in specified lead-free processes. TI may reference these types of products as "Pb-Free".

**RoHS Exempt:** TI defines "RoHS Exempt" to mean products that contain lead but are compliant with EU RoHS pursuant to a specific EU RoHS exemption.

**Green:** TI defines "Green" to mean the content of Chlorine (Cl) and Bromine (Br) based flame retardants meet JS709B low halogen requirements of <=1000ppm threshold. Antimony trioxide based flame retardants must also meet the <=1000ppm threshold requirement.

(3) MSL, Peak Temp. - The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

(4) There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device.

(5) Multiple Device Markings will be inside parentheses. Only one Device Marking contained in parentheses and separated by a "~" will appear on a device. If a line is indented then it is a continuation of the previous line and the two combined represent the entire Device Marking for that device.

(6) Lead/Ball Finish - Orderable Devices may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead/Ball Finish values may wrap to two lines if the finish value exceeds the maximum column width.

**Important Information and Disclaimer:** The information provided on this page represents TI's knowledge and belief as of the date that it is provided. TI bases its knowledge and belief on information provided by third parties, and makes no representation or warranty as to the accuracy of such information. Efforts are underway to better integrate information from third parties. TI has taken and continues to take reasonable steps to provide representative and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. TI and TI suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

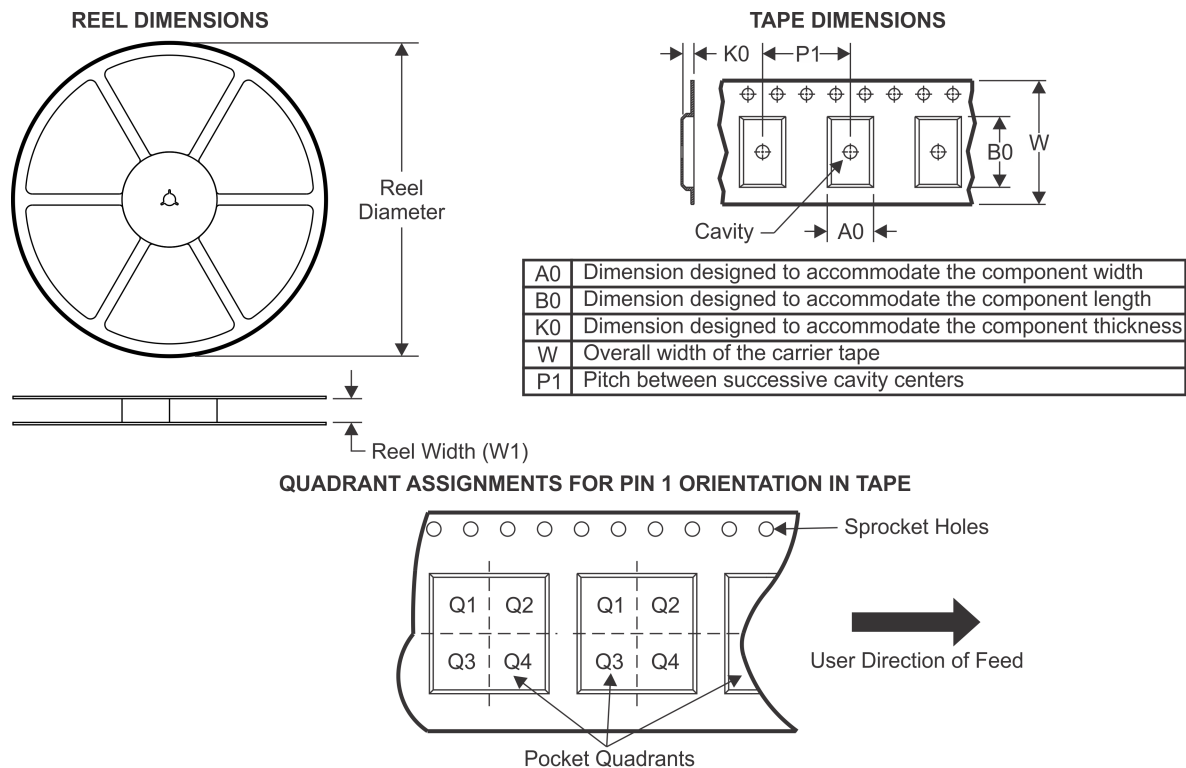
**OTHER QUALIFIED VERSIONS OF SN54AHCT86, SN74AHCT86 :**

- Catalog: [SN74AHCT86](#)

- Military: [SN54AHCT86](#)

NOTE: Qualified Version Definitions:

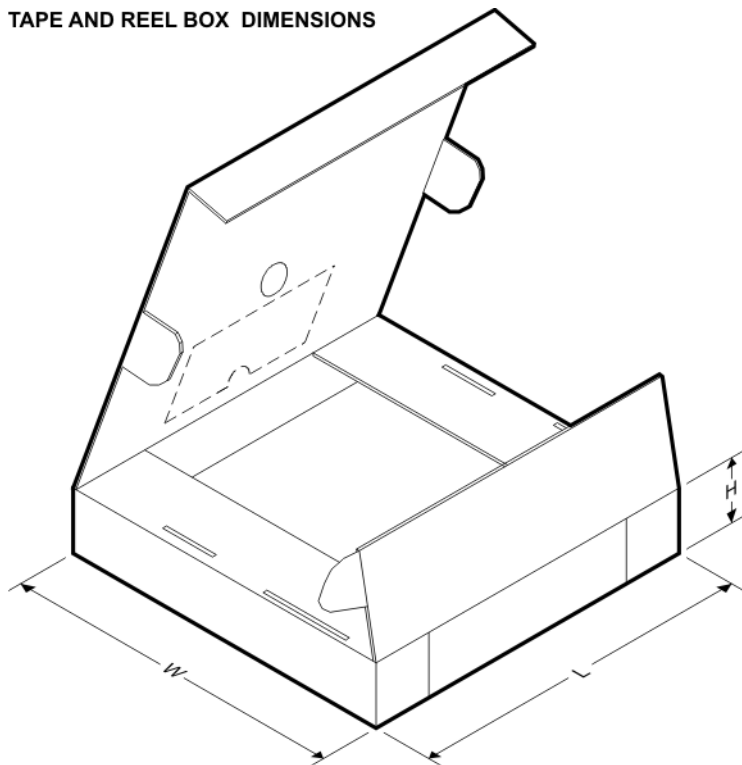
- Catalog - TI's standard catalog product
- Military - QML certified for Military and Defense Applications

**TAPE AND REEL INFORMATION**


\*All dimensions are nominal

| Device         | Package Type | Package Drawing | Pins | SPQ  | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|----------------|--------------|-----------------|------|------|--------------------|--------------------|---------|---------|---------|---------|--------|---------------|
| SN74AHCT86DGVR | TVSOP        | DGV             | 14   | 2000 | 330.0              | 12.4               | 6.8     | 4.0     | 1.6     | 8.0     | 12.0   | Q1            |
| SN74AHCT86DR   | SOIC         | D               | 14   | 2500 | 330.0              | 16.4               | 6.5     | 9.0     | 2.1     | 8.0     | 16.0   | Q1            |
| SN74AHCT86NSR  | SO           | NS              | 14   | 2000 | 330.0              | 16.4               | 8.2     | 10.5    | 2.5     | 12.0    | 16.0   | Q1            |
| SN74AHCT86PWR  | TSSOP        | PW              | 14   | 2000 | 330.0              | 12.4               | 6.9     | 5.6     | 1.6     | 8.0     | 12.0   | Q1            |
| SN74AHCT86RGYR | VQFN         | RGY             | 14   | 3000 | 330.0              | 12.4               | 3.75    | 3.75    | 1.15    | 8.0     | 12.0   | Q1            |

## TAPE AND REEL BOX DIMENSIONS



\*All dimensions are nominal

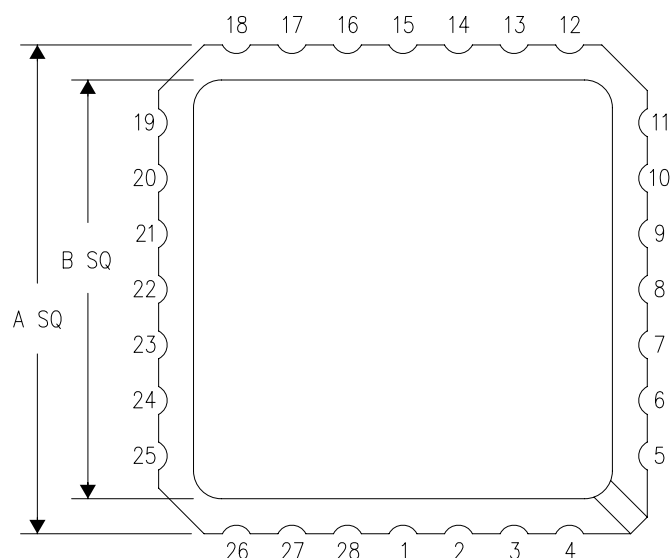
| Device         | Package Type | Package Drawing | Pins | SPQ  | Length (mm) | Width (mm) | Height (mm) |
|----------------|--------------|-----------------|------|------|-------------|------------|-------------|
| SN74AHCT86DGVR | TVSOP        | DGV             | 14   | 2000 | 367.0       | 367.0      | 35.0        |
| SN74AHCT86DR   | SOIC         | D               | 14   | 2500 | 367.0       | 367.0      | 38.0        |
| SN74AHCT86NSR  | SO           | NS              | 14   | 2000 | 367.0       | 367.0      | 38.0        |
| SN74AHCT86PWR  | TSSOP        | PW              | 14   | 2000 | 367.0       | 367.0      | 35.0        |
| SN74AHCT86RGYR | VQFN         | RGY             | 14   | 3000 | 367.0       | 367.0      | 35.0        |



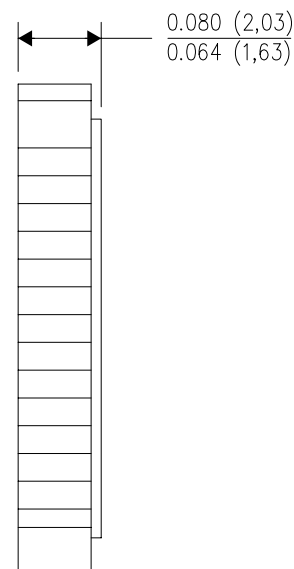
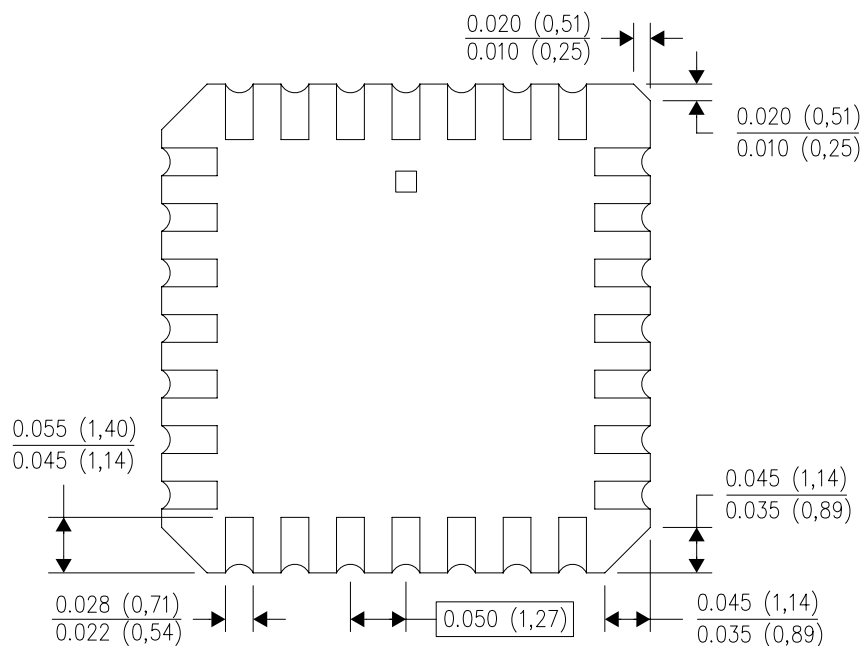
FK (S-CQCC-N\*\*)

LEADLESS CERAMIC CHIP CARRIER

28 TERMINAL SHOWN



| NO. OF<br>TERMINALS<br>** | A                |                  | B                |                  |
|---------------------------|------------------|------------------|------------------|------------------|
|                           | MIN              | MAX              | MIN              | MAX              |
| 20                        | 0.342<br>(8,69)  | 0.358<br>(9,09)  | 0.307<br>(7,80)  | 0.358<br>(9,09)  |
| 28                        | 0.442<br>(11,23) | 0.458<br>(11,63) | 0.406<br>(10,31) | 0.458<br>(11,63) |
| 44                        | 0.640<br>(16,26) | 0.660<br>(16,76) | 0.495<br>(12,58) | 0.560<br>(14,22) |
| 52                        | 0.740<br>(18,78) | 0.761<br>(19,32) | 0.495<br>(12,58) | 0.560<br>(14,22) |
| 68                        | 0.938<br>(23,83) | 0.962<br>(24,43) | 0.850<br>(21,6)  | 0.858<br>(21,8)  |
| 84                        | 1.141<br>(28,99) | 1.165<br>(29,59) | 1.047<br>(26,6)  | 1.063<br>(27,0)  |

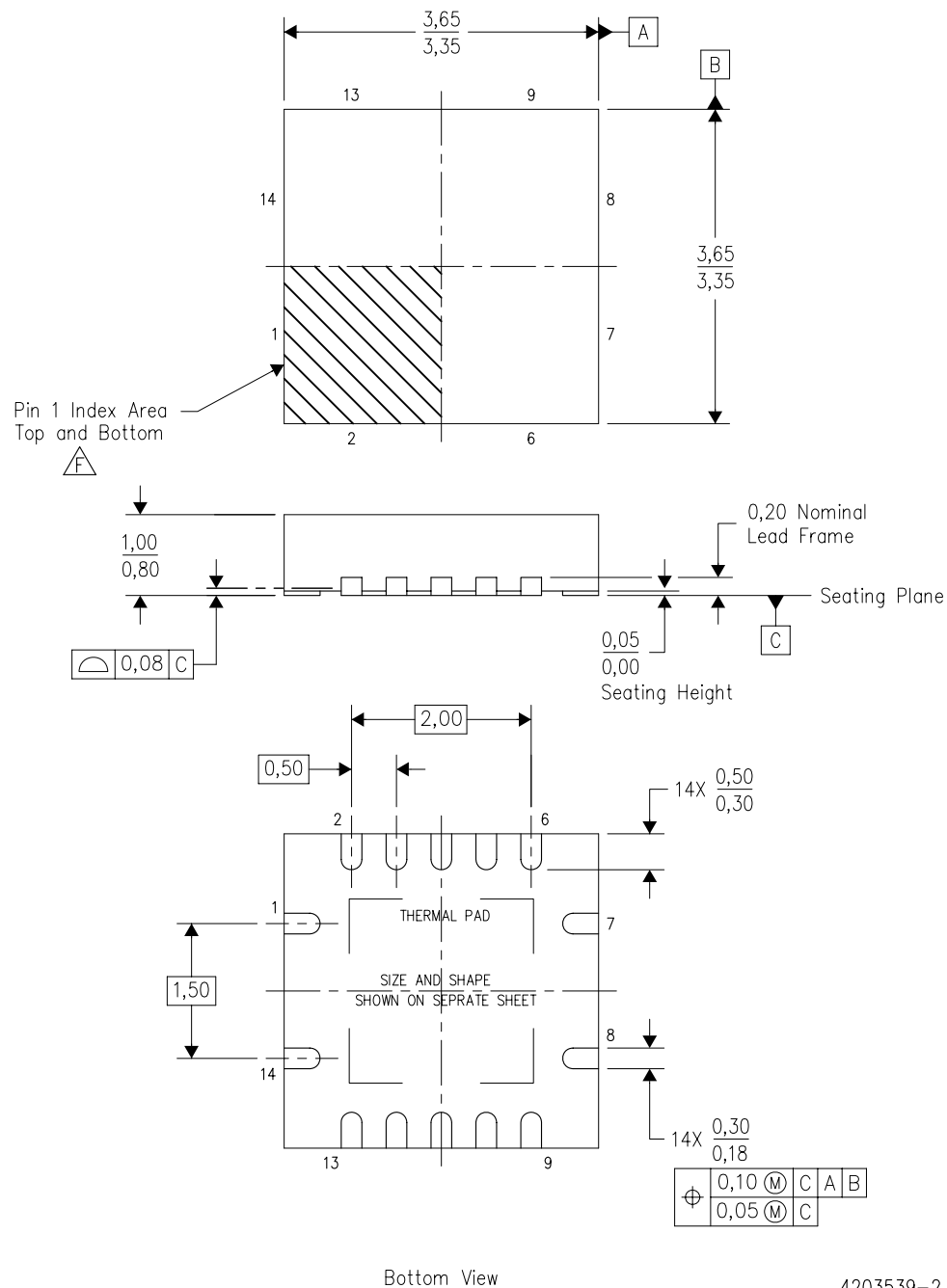


4040140/D 01/11

- NOTES:
- All linear dimensions are in inches (millimeters).
  - This drawing is subject to change without notice.
  - This package can be hermetically sealed with a metal lid.
  - Falls within JEDEC MS-004

RGY (S-PVQFN-N14)

PLASTIC QUAD FLATPACK NO-LEAD



4203539-2/I 06/2011

- NOTES:
- All linear dimensions are in millimeters. Dimensioning and tolerancing per ASME Y14.5M-1994.
  - This drawing is subject to change without notice.
  - QFN (Quad Flatpack No-Lead) package configuration.
  - The package thermal pad must be soldered to the board for thermal and mechanical performance.
  - See the additional figure in the Product Data Sheet for details regarding the exposed thermal pad features and dimensions.
  - Pin 1 identifiers are located on both top and bottom of the package and within the zone indicated. The Pin 1 identifiers are either a molded, marked, or metal feature.
  - Package complies to JEDEC MO-241 variation BA.

RGY (S-PVQFN-N14)

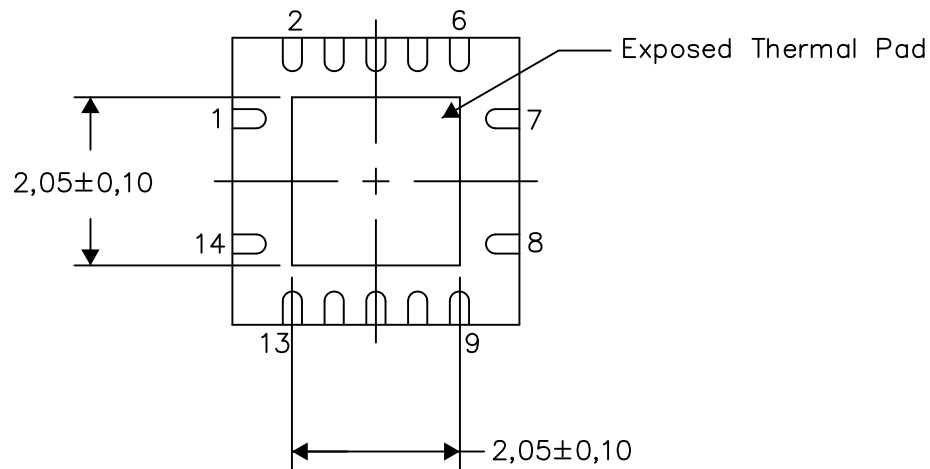
PLASTIC QUAD FLATPACK NO-LEAD

## THERMAL INFORMATION

This package incorporates an exposed thermal pad that is designed to be attached directly to an external heatsink. The thermal pad must be soldered directly to the printed circuit board (PCB). After soldering, the PCB can be used as a heatsink. In addition, through the use of thermal vias, the thermal pad can be attached directly to the appropriate copper plane shown in the electrical schematic for the device, or alternatively, can be attached to a special heatsink structure designed into the PCB. This design optimizes the heat transfer from the integrated circuit (IC).

For information on the Quad Flatpack No-Lead (QFN) package and its advantages, refer to Application Report, QFN/SON PCB Attachment, Texas Instruments Literature No. SLUA271. This document is available at [www.ti.com](http://www.ti.com).

The exposed thermal pad dimensions for this package are shown in the following illustration.



Bottom View

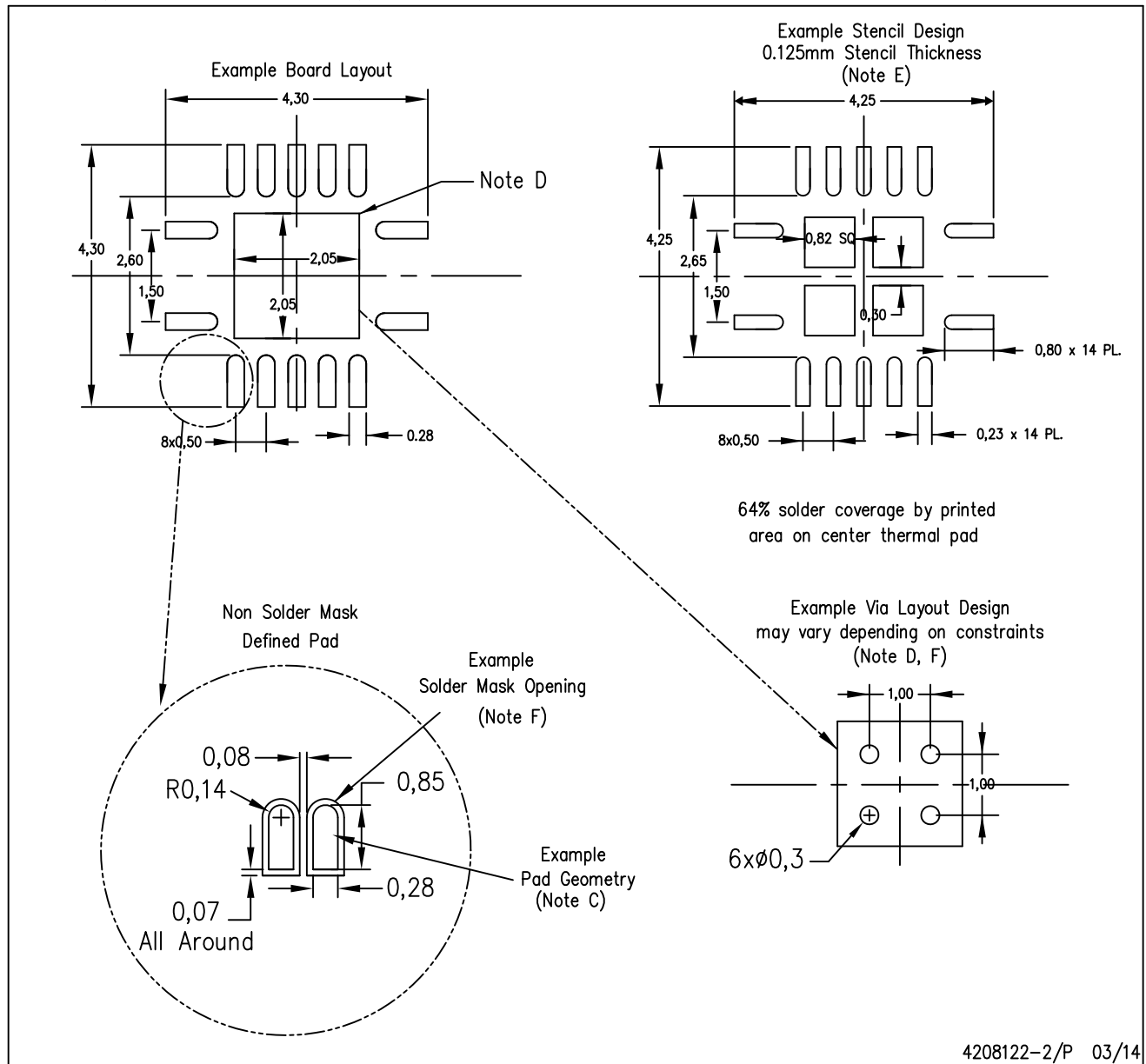
Exposed Thermal Pad Dimensions

4206353-2/P 03/14

NOTE: All linear dimensions are in millimeters

RGY (S-PVQFN-N14)

PLASTIC QUAD FLATPACK NO-LEAD



4208122-2/P 03/14

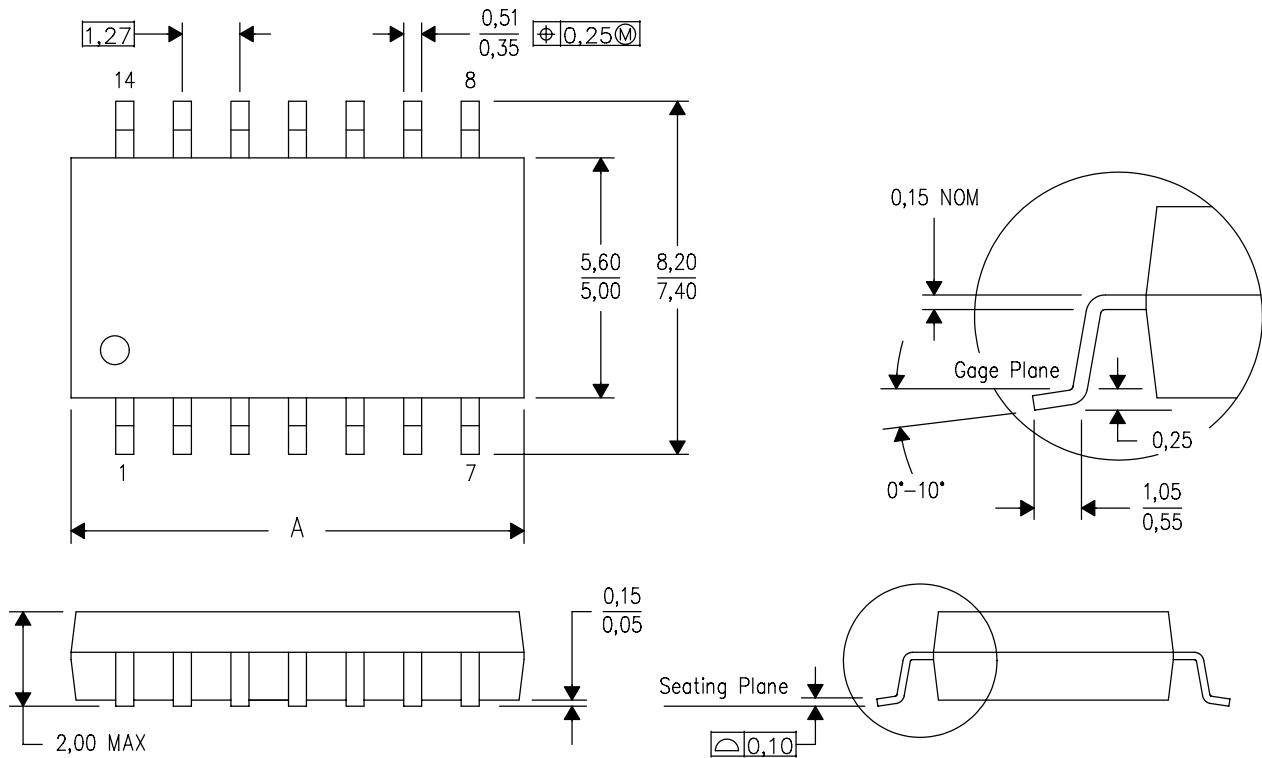
- NOTES:
- All linear dimensions are in millimeters.
  - This drawing is subject to change without notice.
  - Publication IPC-7351 is recommended for alternate designs.
  - This package is designed to be soldered to a thermal pad on the board. Refer to Application Note, Quad Flat-Pack QFN/SON PCB Attachment, Texas Instruments Literature No. SLUA271, and also the Product Data Sheets for specific thermal information, via requirements, and recommended board layout. These documents are available at [www.ti.com](http://www.ti.com) <<http://www.ti.com>>.
  - Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Refer to IPC 7525 for stencil design considerations.
  - Customers should contact their board fabrication site for minimum solder mask web tolerances between signal pads.

# MECHANICAL DATA

NS (R-PDSO-G\*\*)

PLASTIC SMALL-OUTLINE PACKAGE

14-PINS SHOWN



| DIM \ PINS ** | 14    | 16    | 20    | 24    |
|---------------|-------|-------|-------|-------|
| A MAX         | 10,50 | 10,50 | 12,90 | 15,30 |
| A MIN         | 9,90  | 9,90  | 12,30 | 14,70 |

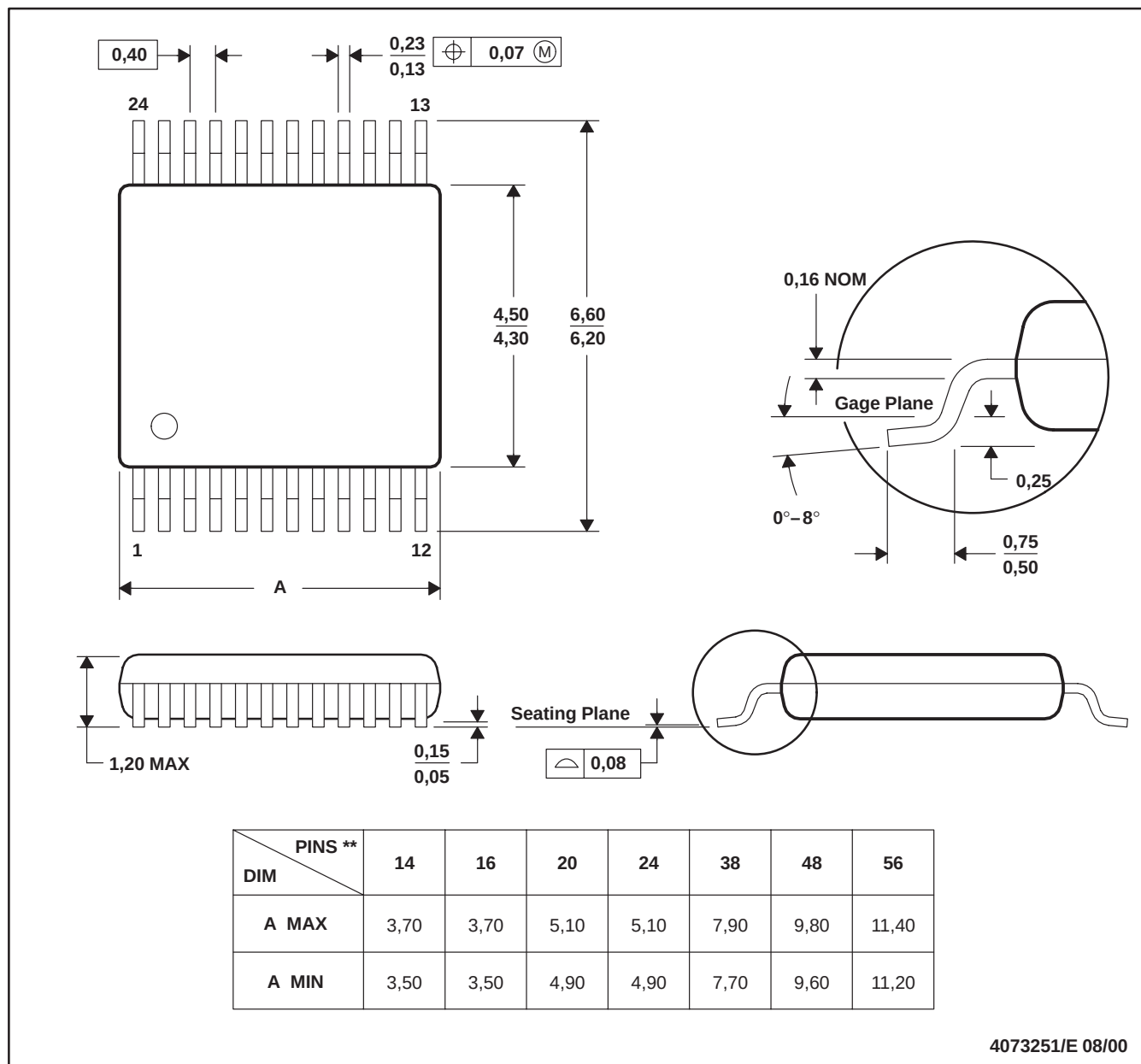
4040062/C 03/03

- NOTES:
- A. All linear dimensions are in millimeters.
  - B. This drawing is subject to change without notice.
  - C. Body dimensions do not include mold flash or protrusion, not to exceed 0,15.

## DGV (R-PDSO-G\*\*)

## PLASTIC SMALL-OUTLINE

24 PINS SHOWN



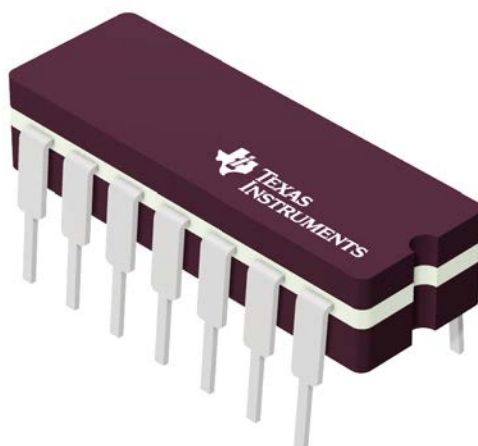
- NOTES: A. All linear dimensions are in millimeters.  
 B. This drawing is subject to change without notice.  
 C. Body dimensions do not include mold flash or protrusion, not to exceed 0,15 per side.  
 D. Falls within JEDEC: 24/48 Pins – MO-153  
 14/16/20/56 Pins – MO-194

**J 14**

## GENERIC PACKAGE VIEW

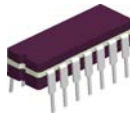
**CDIP - 5.08 mm max height**

CERAMIC DUAL IN LINE PACKAGE

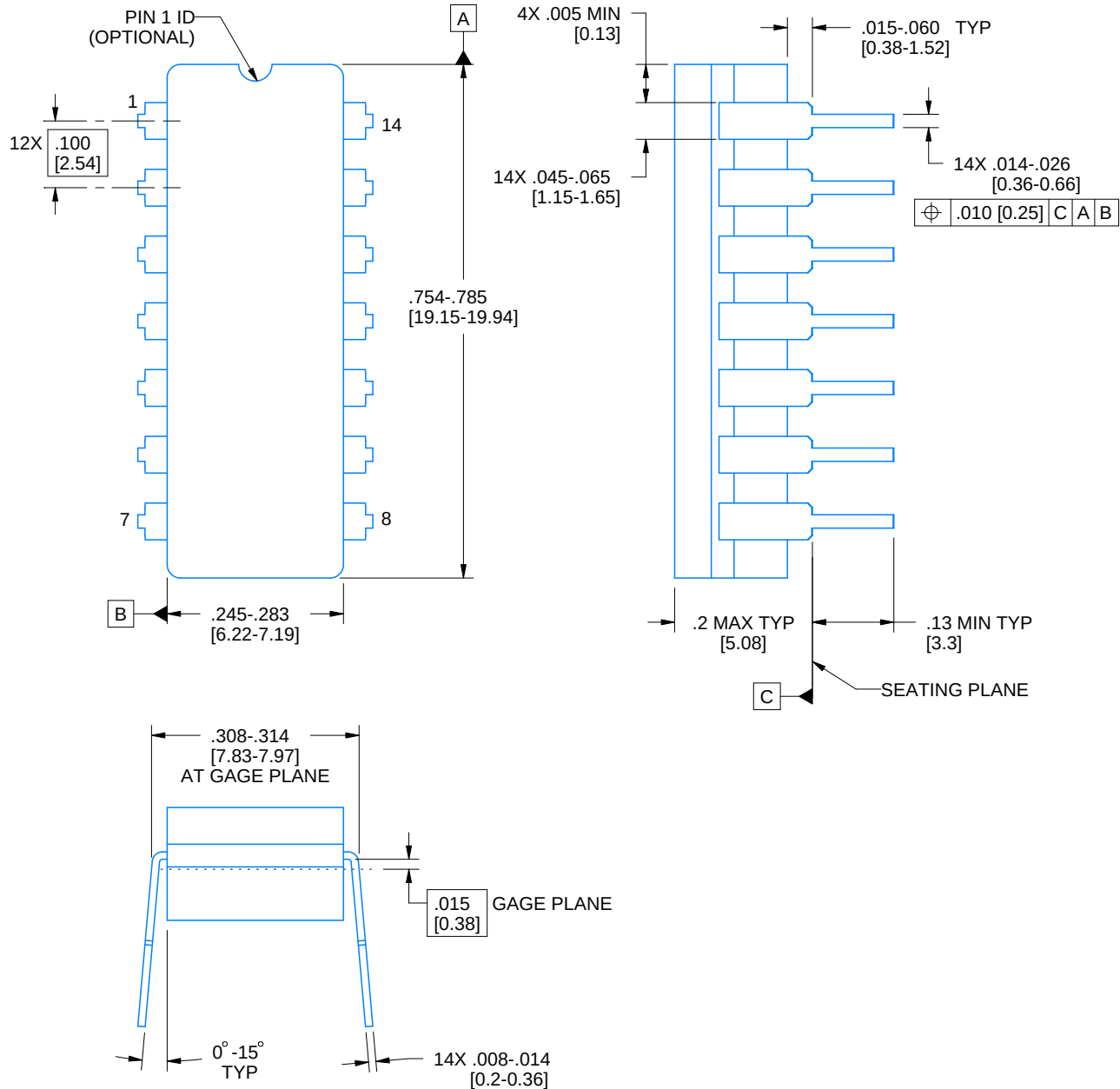


Images above are just a representation of the package family, actual package may vary.  
Refer to the product data sheet for package details.

4040083-5/G

**J0014A****PACKAGE OUTLINE****CDIP - 5.08 mm max height**

CERAMIC DUAL IN LINE PACKAGE



4214771/A 05/2017

**NOTES:**

1. All controlling linear dimensions are in inches. Dimensions in brackets are in millimeters. Any dimension in brackets or parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This package is hermetically sealed with a ceramic lid using glass frit.
4. Index point is provided on cap for terminal identification only and on press ceramic glass frit seal only.
5. Falls within MIL-STD-1835 and GDIP1-T14.



**TEXAS  
INSTRUMENTS**  
www.ti.com

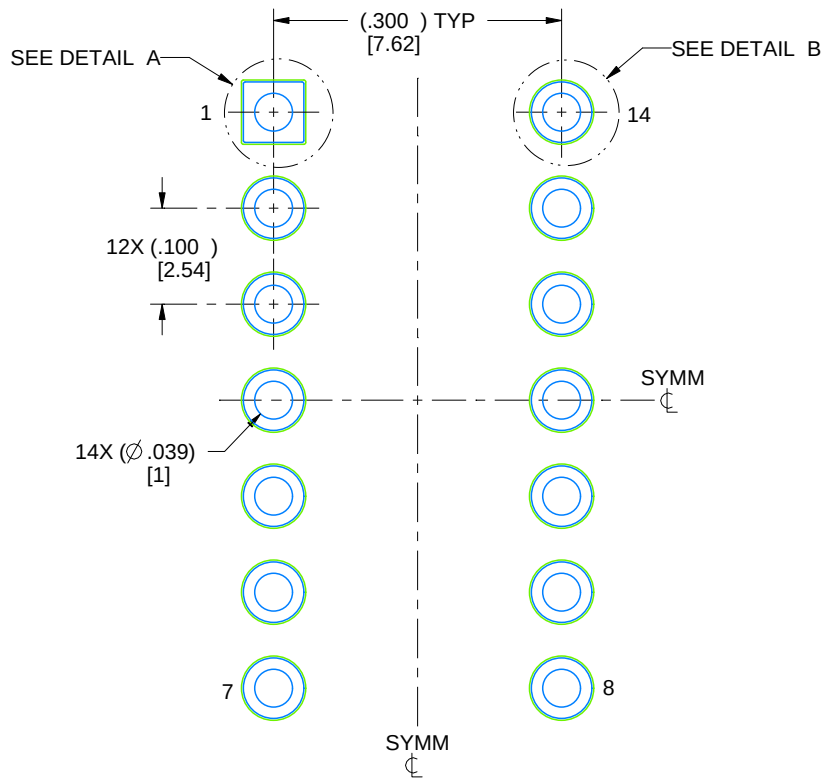


# EXAMPLE BOARD LAYOUT

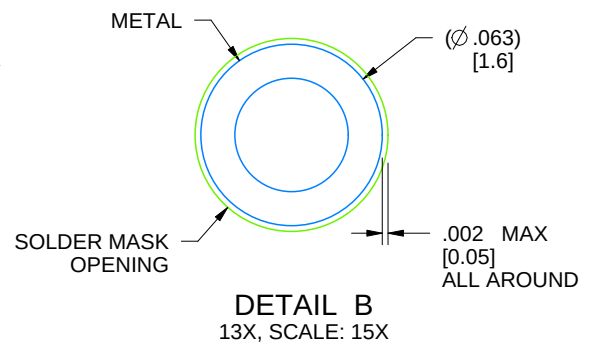
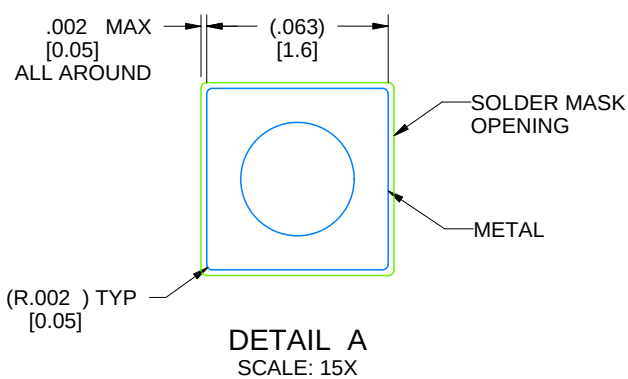
J0014A

CDIP - 5.08 mm max height

CERAMIC DUAL IN LINE PACKAGE



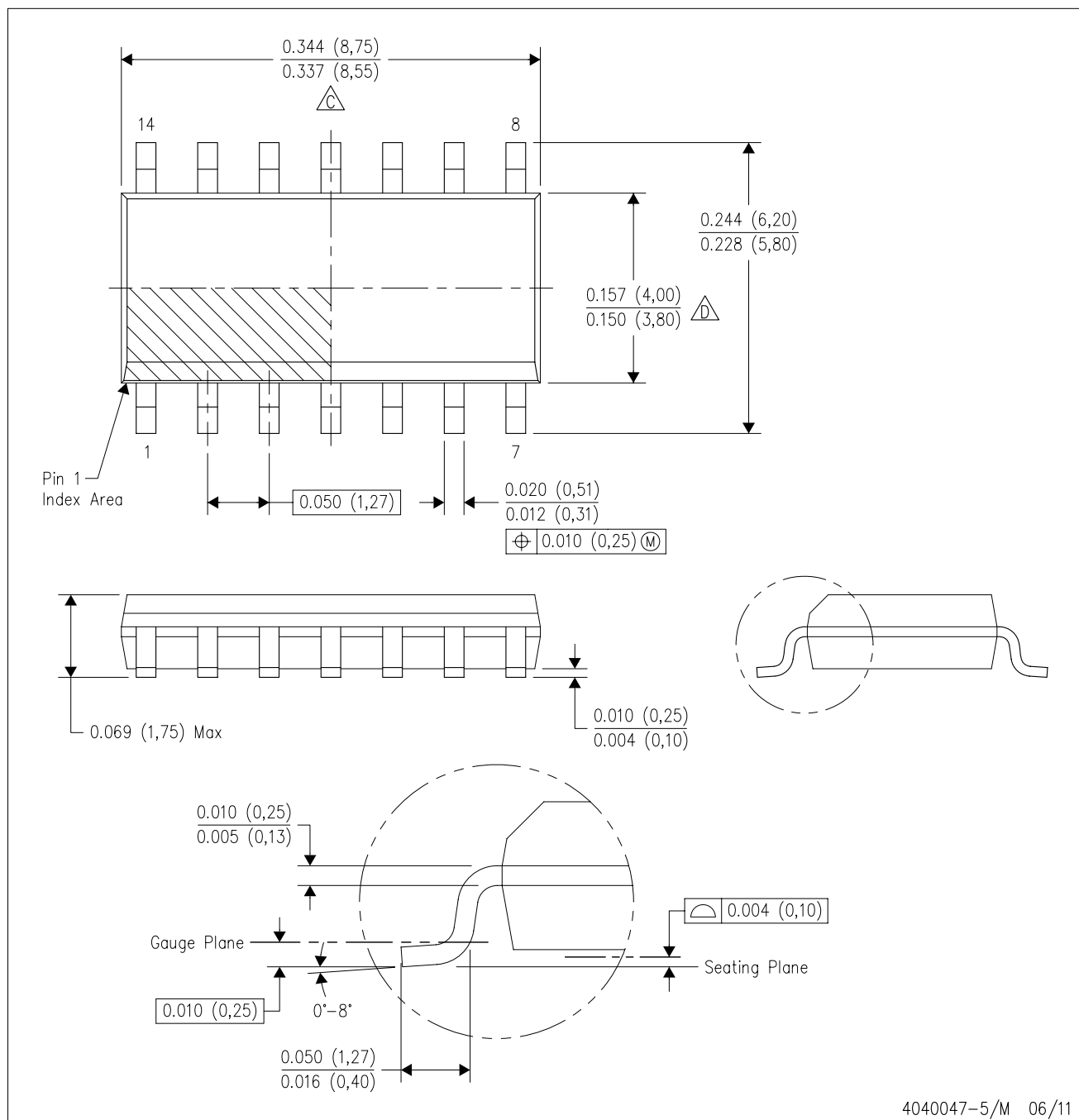
LAND PATTERN EXAMPLE  
NON-SOLDER MASK DEFINED  
SCALE: 5X



4214771/A 05/2017

D (R-PDSO-G14)

PLASTIC SMALL OUTLINE



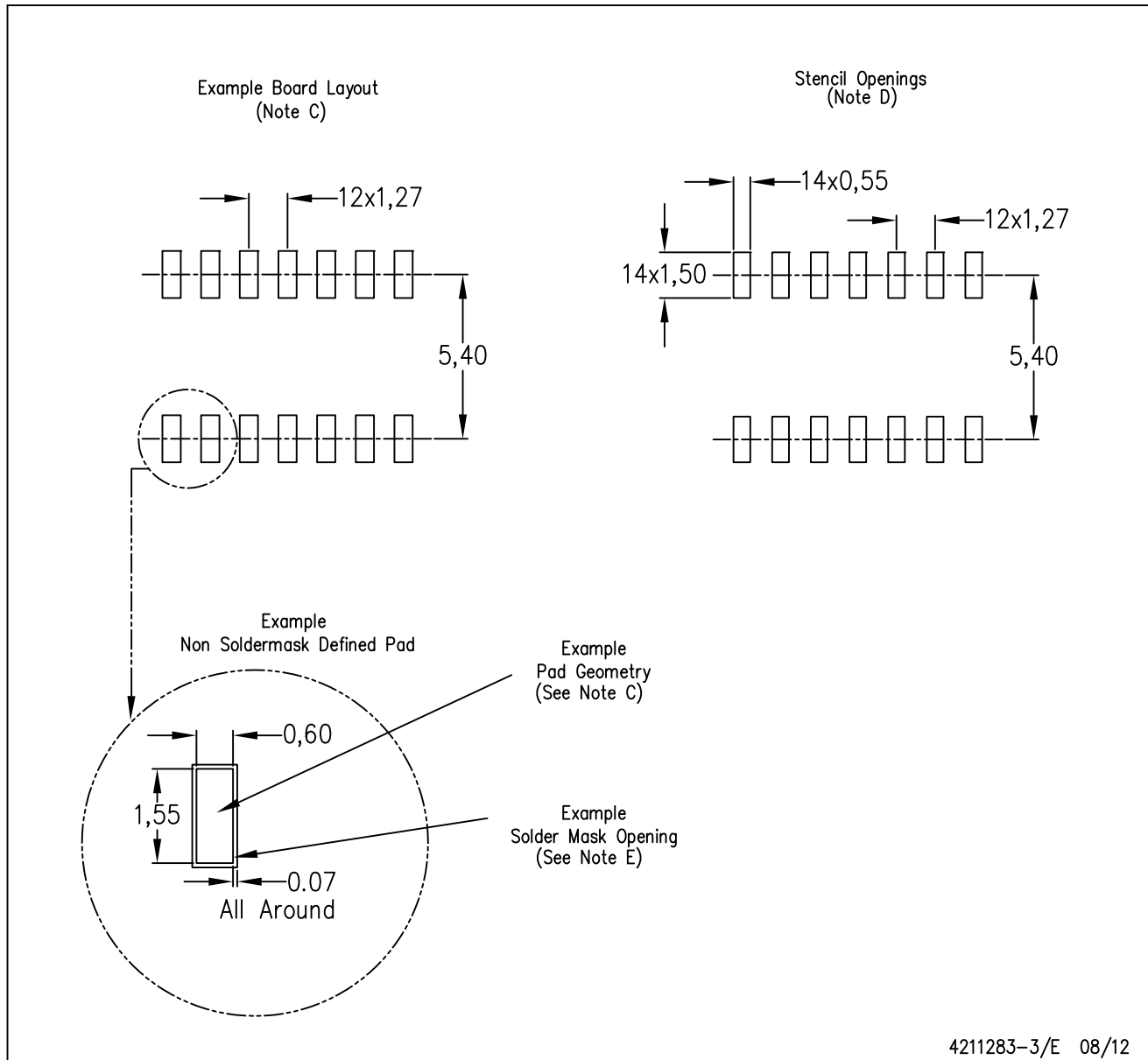
4040047-5/M 06/11

NOTES:

- A. All linear dimensions are in inches (millimeters).
- B. This drawing is subject to change without notice.
- C. Body length does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.006 (0,15) each side.
- D. Body width does not include interlead flash. Interlead flash shall not exceed 0.017 (0,43) each side.
- E. Reference JEDEC MS-012 variation AB.

D (R-PDSO-G14)

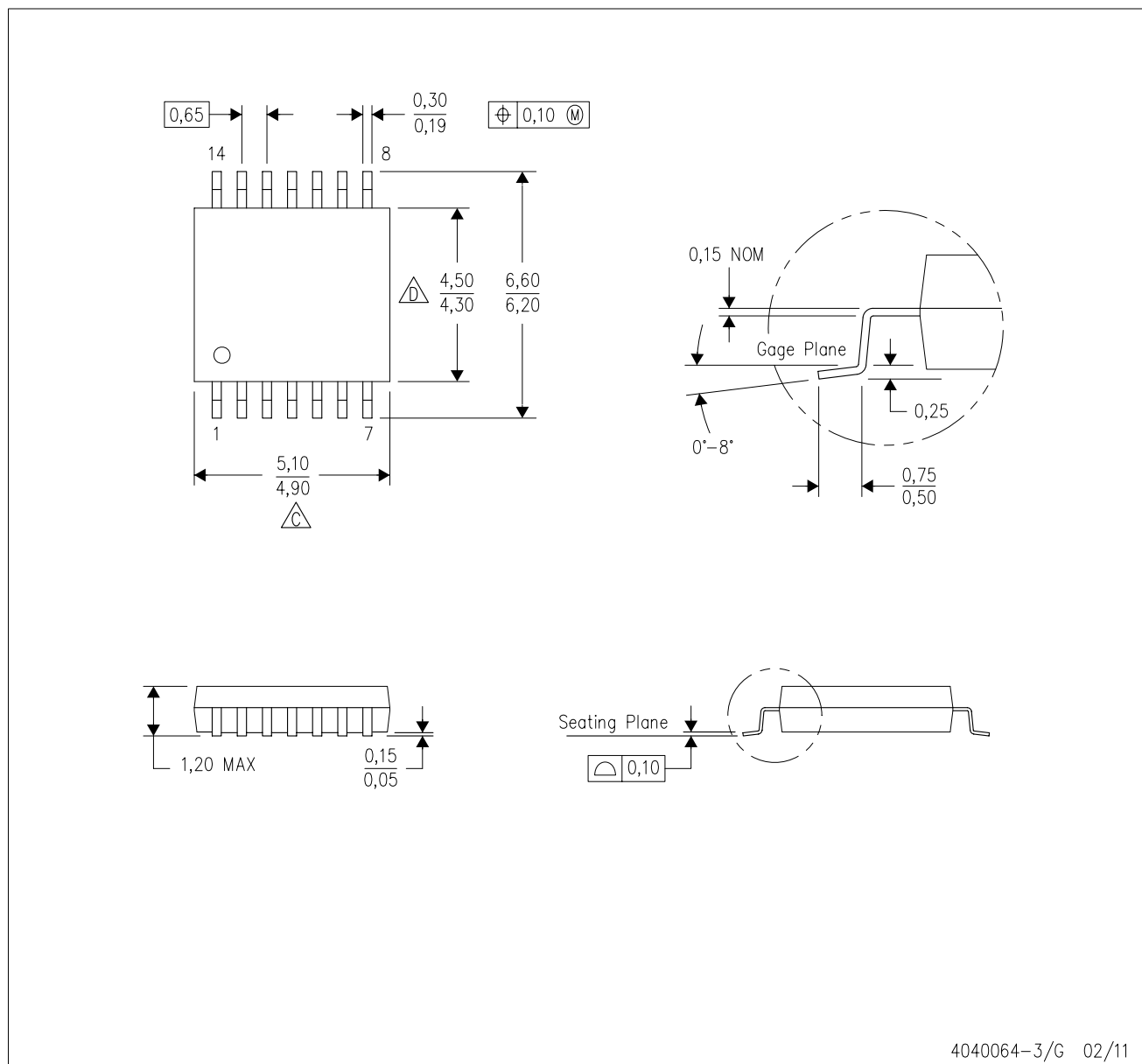
PLASTIC SMALL OUTLINE



- NOTES:
- A. All linear dimensions are in millimeters.
  - B. This drawing is subject to change without notice.
  - C. Publication IPC-7351 is recommended for alternate designs.
  - D. Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Refer to IPC-7525 for other stencil recommendations.
  - E. Customers should contact their board fabrication site for solder mask tolerances between and around signal pads.

PW (R-PDSO-G14)

PLASTIC SMALL OUTLINE



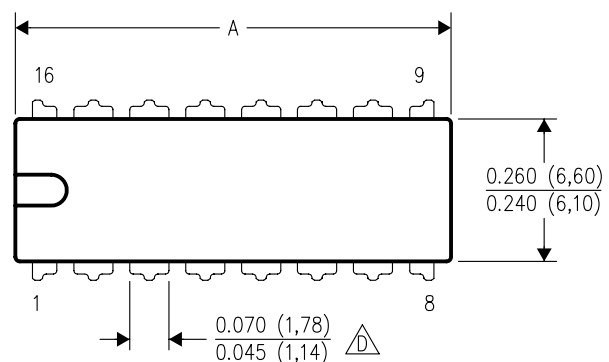
4040064-3/G 02/11

- NOTES:
- A. All linear dimensions are in millimeters. Dimensioning and tolerancing per ASME Y14.5M-1994.
  - B. This drawing is subject to change without notice.
  - C. Body length does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0,15 each side.
  - D. Body width does not include interlead flash. Interlead flash shall not exceed 0,25 each side.
  - E. Falls within JEDEC MO-153

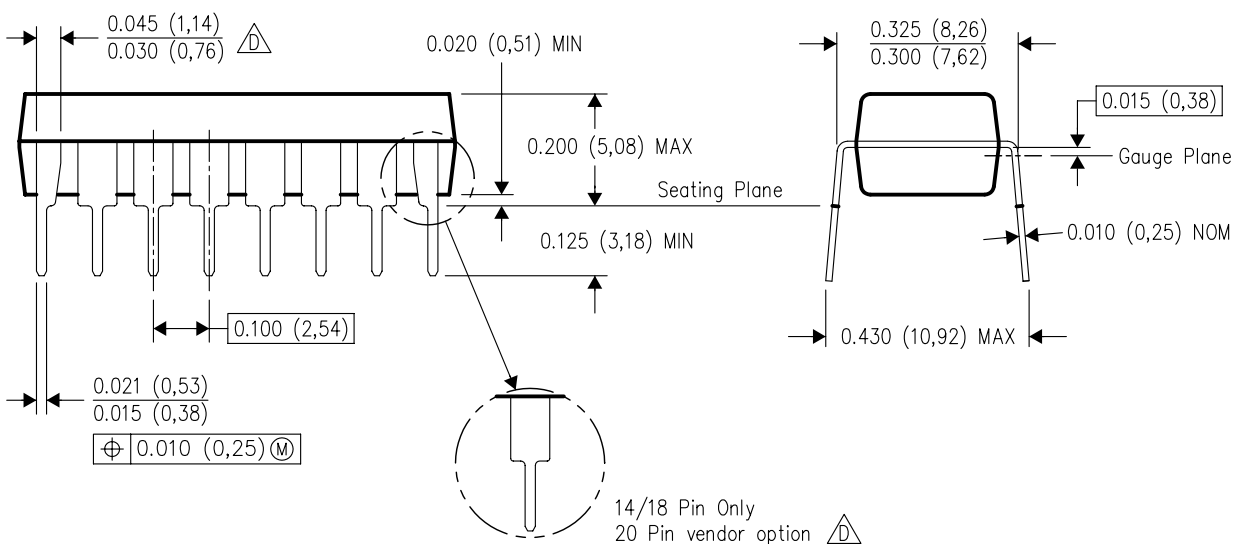
N (R-PDIP-T\*\*)

16 PINS SHOWN

## PLASTIC DUAL-IN-LINE PACKAGE



| PINS **<br>DIM      | 14               | 16               | 18               | 20               |
|---------------------|------------------|------------------|------------------|------------------|
| A MAX               | 0.775<br>(19,69) | 0.775<br>(19,69) | 0.920<br>(23,37) | 1.060<br>(26,92) |
| A MIN               | 0.745<br>(18,92) | 0.745<br>(18,92) | 0.850<br>(21,59) | 0.940<br>(23,88) |
| MS-001<br>VARIATION | AA               | BB               | AC               | AD               |



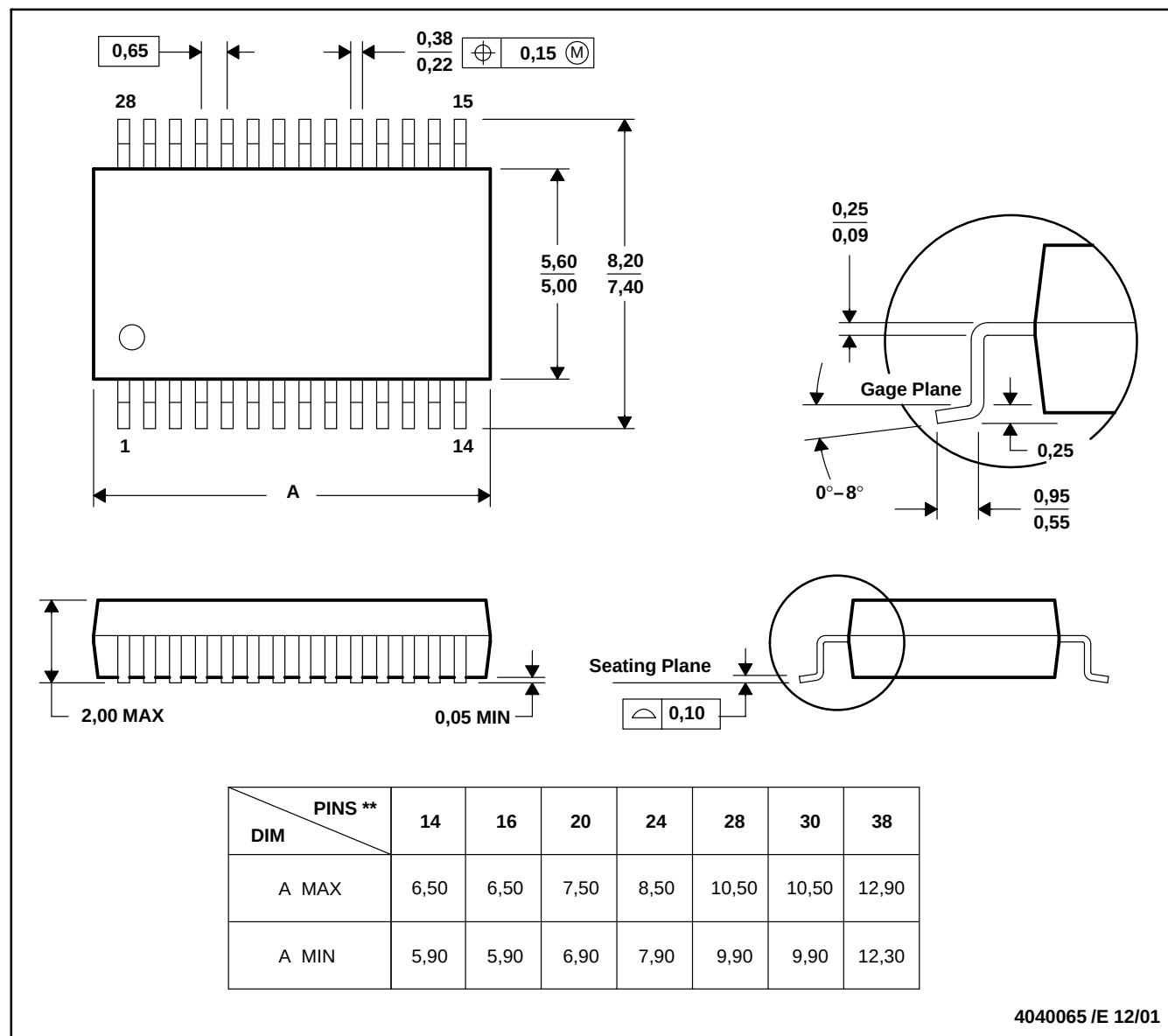
4040049/E 12/2002

- NOTES:
- A. All linear dimensions are in inches (millimeters).
  - B. This drawing is subject to change without notice.
  -  Falls within JEDEC MS-001, except 18 and 20 pin minimum body length (Dim A).
  -  The 20 pin end lead shoulder width is a vendor option, either half or full width.

## DB (R-PDSO-G\*\*)

## PLASTIC SMALL-OUTLINE

28 PINS SHOWN



- NOTES: A. All linear dimensions are in millimeters.  
 B. This drawing is subject to change without notice.  
 C. Body dimensions do not include mold flash or protrusion not to exceed 0,15.  
 D. Falls within JEDEC MO-150

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